

Niveditha B Menon

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EDUCATION

- Vellore Institute of Technology** Vellore, Tamil Nadu, India
 - Bachelor of Technology - Mechanical Engineering; CGPA: 8.73/10.00*
Expected May 2027
 - Minor: Computer Science Engineering.*

SKILLS SUMMARY

- Languages:** Python, Java
- Engineering Software:** Solidworks, ANSYS, MATLAB, AutoCAD, Fusion360.
- Other Software and Skills:** Microsoft Office.

RELEVANT COURSEWORK

- Engineering Mechanics, Calculus, Engineering Physics, Engineering Design and Visualization (Solidworks, Fusion 360), Thermodynamics, Differential Equations and Transforms, Mechanics of Solids, Fluid Mechanics, Complex Numbers and Linear Algebra, Material Science, Macroeconomics

RESEARCH, AND INDUSTRIAL EXPERIENCE

- Fertilizers And Chemicals Travancore (FACT), Kochi, Kerala, India** Engineering Intern
December 2024
 - Fertilizer Plant Study*
 - Gained hands-on experience in industrial operations through plant visits, and machinery analysis.
 - Developed expertise in the functionality and maintenance of equipment, including compressors, dryers, granulators, valves, gates, and other mechanical systems critical to fertilizer manufacturing.
 - Enhanced technical knowledge, and problem-solving skills by observing real-world applications of industrial processes, and engineering concepts.
 - Ensured safety, and efficiency during plant operations in a dynamic industrial environment.
- Vellore Institute of Technology, Vellore, Tamil Nadu, India** Undergraduate Research Assistant
July 2024 - November 2024
 - Lithium Ferrous Phosphate Battery Study*
 - Investigated the performance, and health assessment of a lithium ferrous phosphate (LFP) battery pack consisting of 23 cells connected in series balancing mode.
 - Conducted experiments at various current ranges (3A, 5A, 8A) using a custom-developed battery management system to estimate optimized performance and overall health conditions.
 - Analyzed LFP battery pack characteristics at different power modes (20W-750W) to assess charging efficiency and performance.
 - Recorded a 58% State Of Charge (SOC) capability within a 3.3-hour runway period, with full charging achieved after reaching 50% SOC under all current modes.
 - Observed a high State Of Health (SOH) at 92% throughout the investigation, indicating robust thermal stability under temperature variations of 26°C to 31°C.
 - Concluded that LFP battery packs are highly competitive for energy storage applications in on-road, and off-road transportation due to their superior performance, and thermal reliability.
 - Culminated in the publication of a conference proceeding, showcasing the findings, and their relevance to energy storage and electric vehicle applications.
- Vellore Institute of Technology, Vellore, Tamil Nadu, India** Undergraduate Research Assistant
July 2024 - Present
 - Thermal Management of Lithium Ferrous Phosphate Battery*
 - Researching solutions to address battery overheating issues to extend battery lifespan.
 - Conducting MATLAB simulations to model and analyze battery temperatures under various conditions.
 - Performing a comprehensive literature survey of 35-40 papers to understand the current state-of-the-art in battery thermal management.

PRESENTATIONS AND PUBLICATIONS

- B Menon Niveditha, K Nanthagopal.** (2024). Assessment of Lithium Ferrous Phosphate Battery Cells Under Series Balancing Mode-Performance and Health Behaviours, ICOFT MADE 4.0, NIT Puducherry, Tamil Nadu, India.

EXTRA CURRICULAR ACTIVITIES

- Designed, and manufactured a fixed-wing plane and participated in SAE VIT Aerodominator 2024 as an aeronautics engineering member of the Curtiss R/C club.
- Member of Toastmasters International public speaking club during high school, developing communication, and leadership skills.
- Hosted a Formula One racing quiz competition under SAE, showcasing event organization, and subject matter expertise.